Hans-Peter Marshall

Cryosphere Geophysics And Remote Sensing (CryoGARS) group
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Education

2005	Ph.D, Civil Engineering (Geotechnical), GPA: 3.9/4.0 Institute of Arctic and Alpine Research, University of Colorado at Boulder Thesis: *Snowpack spatial variability: towards understanding its effect on remote sensing measurements and snow slope stability*
1999	B.S., Physics with honors, Geophysics Minor (1^{st} in UW history), GPA: 3.6/4.0 University of Washington Thesis: *Wet snow densification*
1994	Nathan Hale High School, Seattle, Washington, GPA: 3.9/4.0

Selected Professional Experience

08/2018-8/2021	Project Scientist, NASA SnowEx Mission
08/2016-07/2018	Co-lead of Field Campaign, NASA SnowEx 2017 Mission
08/2013-present	Associate Professor, Department of Geosciences and Center for Geophysical Investigation of the Shallow Subsurface (CGISS), Boise State University
07/2007-05/2020	Geophysical Engineer (Expert Consultant), U.S. Army Corps of Engineers Cold Regions Research and Engineering Lab (CRREL), Hanover, New Hampshire
08/2008-07/2013	Assistant Professor, Professor, Department of Geosciences and Center for Geophysical Investigation of the Shallow Subsurface (CGISS), Boise State University

Research Interests

Snow science and glaciology, using geophysics, engineering, and remote sensing.
Spatial variability of snow, snow and ice mechanics, snow slope stability, meltwater dynamics
Computational statistics, geostatistics, compressed sensing, instrumentation development

Selected Honors and Awards

2020	BSU Faculty Excellence Award
2019	NASA Instrument Incubator Program Award, NASA, 1st in Idaho history
2019	2018 WRR Editors' Choice Award: Hedrick et al., 2018
2018	NASA Robert H. Goddard Team Award: SnowEx Year 1 Organizing Team
2012	Timeless Award, University of Washington, 1 of 150 College of Arts and Sciences alumni awarded from UW history on 150th yr anniv. (1861-2011), 1 of 3 from Physics Department
2010-13	New Investigator Program Award, NASA, 1st in Idaho history
2010	Young Scientist Award, Cryosphere Section, American Geophysical Union
2009	Top Reviewer Award, Cold Regions Science and Technology (CRST)
2008	U.S. Young Scientist Award, International Union of Radio Science (URSI)

Selected Peer-Reviewed Publications [total=67, h-index=27, i10-index=50, citations=2272]

2022 Lievens, H., Brangers, I., **H.-P. Marshall**, Jonas, T., Olefs, M., and DeLannoy, G. (2022). Sentinel-1 snow depth retrieval at sub-kilometer resolution over the European Alps. *The Cryosphere*, 16(1):159–177, doi.org/10.5194/tc–16–159–2022

2021 **H.-P. Marshall**, Deeb, E., Forster, R., Vuyovich, C., Elder, K., Hiemstra, C., and Lund, J. (2021). L-band InSAR depth retrieval during the NASA SnowEx 2020 Campaign: Grand Mesa, Colorado. *Proceedings of the International Geoscience and Remote Sensing Symposium (IGARSS)*, pages 625–627, DOI:10.1109/IGARSS47720.2021.9553852

Johnson, J., Anderson, J., **H.-P. Marshall**, Havens, S., and Watson, L. (2021). Snow avalanche detection and source constraints made using multiple infrasound sensor arrays. *Journal of Geophysical Research-Earth Surface*, 126(3):e2020JF005741

Meehan, T. G., **H.P. Marshall**, Bradford, J. H., Hawley, R. L., Overly, T. B., Lewis, G., Graeter, K., Osterburg, E., and McCarthy, F. (2021). Historical surface mass balance reconstruction 1984-2017 from greentracs multi-offset ground-penetrating radar. *Journal of Glaciology*, 67(262):219–228, doi:10.1017/jog.2020.91

Lievens, H., Brangers, I., **H.-P. Marshall**, Jonas, T., Olefs, M., and DeLannoy, G. (2022). Sentinel-1 snow depth retrieval at sub-kilometer resolution over the European Alps. *The Cryosphere*, 16(1):159–177, doi.org/10.5194/tc–16–159–2022

Lund, J., Forster, R. R., Rupper, S. B., **H.P. Marshall**, Deeb, E. J., and Hashmi, M. (2020). Mapping snowmelt progression in the upper indus basin with synthetic aperture radar. *Frontiers in Earth Science*, 7:318, doi:10.3389/feart.2019.00318

2019 Lievens, H., Demuzere, M., **H.P. Marshall**, Reichle, R. H., Brucker, L., Brangers, I., de Rosnay, P., Dumont, M., Girotto, M., Immerzeel, W. W., Jonas, T., Kim, E. J., Koch, I., Marty, C., Saloranta, T., Schöber, J., and Lannoy, G. J. D. (2019). Snow depth variability in the northern hemisphere mountains observed from space. *Nature Communications*, 10(1):1–12

McGrath, D., Webb, R., Shean, D., Bonnell, R., **H.P. Marshall**, Painter, T., Molotch, N., Elder, K., Hiemstra, C., and Brucker, L. (2019). Spatially extensive ground-penetrating radar snow depth observations during NASA's 2017 SnowEx Campaign: Comparison to in situ, airborne, and satellite observations. *Water Resources Research*, 55(11):10026–10036, doi:10.1029/2019WR024907

2018 Hedrick, A., Marks, D., Kormos, P., **H.P. Marshall**, Havens, S., Bormann, K., and Painter, T. H. (2018). Direct insertion of NASA Airborne Snow Observatory-derived snow depth time-series into the iSnobal energy balance snow model. *Water Resources Research*, 54(10):8045–8063, doi.org/10.1029/2018WR023190

Kim, Y., Reck, T. J., Alonso-delPino, M., Painter, T. H., **H.P. Marshall**, Bair, E. H., Dozier, J., Chattopadhyay, G., Liou, K.-N., Chang, M.-C. F., and Tang, A. (2018). A Ku-band CMOS FMCW radar transceiver for snowpack remote sensing. *IEEE Transactions On Microwave Theory and Techniques*, 66(5):2480–2494